Tyler Rydell

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Education

Bergen Community College — Paramus, NJ
Associate in Science, Engineering Science — *September 2025 – May 2027*
Cumulative GPA: 4.0 / 4.0 | Judith K. Winn School of Honors | Phi Theta Kappa | Dean's List (2x)

Experience

Prototype Engineer / Lab Technician — July 2026 – September 2026
Flores Engineering — *Kearny, NJ*
- Designed and iterated mechanical parts in Fusion 360, then fabricated and assembled prototypes in the shop
- Assembled and tested full prototypes, troubleshooting across mechanical systems, electrical systems, and code
- Operated and maintained FDM and SLA 3D printers, supporting in-shop fabrication
- Developed internal tooling and structured multi-variant BOMs to streamline inventory and part sourcing

Summer Research Intern — May 2026 – August 2026
STEM Research Center at Bergen Community College — *Paramus, NJ*
- Delivered a prosthetic arm prototype and research poster with a six-member team, winning first place of 42 teams at the STEM Research Center poster competition
- Owned embedded C++ firmware, circuit design, and most mechanical design while coordinating sub-teams
- This work has since grown into BARAA, an ongoing benchmark for prosthetic and humanoid arms, presented at NERC 2026 at Princeton

Projects

ARMin: A Low-Cost Tendon-Driven Prosthetic Arm | *Onshape, C++, Python* — October 2025 – August 2026
- Designed a 12-DOF prosthetic arm for under $300 per unit versus commercial myoelectric hands
- Engineered tendon-driven fingers, direct-drive splay, and a dual-planetary-gearbox elbow (64:1 and 4:1)
- Fully 3D printed the structure in PLA and PETG-CF

LiDAR Alignment Mount | *Onshape* — September 2026 – Present
- Designed a LiDAR mount for Dr. Joseph Sivo's research team to measure drag coefficients of falling objects
- Added collimation-screw alignment lasers and spring-loaded mounting screws for two-stage alignment
- Housed the electronics inside the mount with a panel-mount barrel jack, push buttons, and micro USB

Animatronic Head | *Onshape, C++* — August 2026
- Built a 2-DOF neck and 2-DOF eyes driven by servo pushrod linkages, run by joystick or autonomously

Leadership

President — May 2026 – Present
Bergen Robotics — *Bergen Community College*
- Promoted to team lead by the coach after contributing to programming and CAD through a VEX Worlds run
- Coordinated design and programming sub-teams, keeping a stable shared codebase through robot iterations
- Increased team membership from 10 to 30 members

President — December 2025 – Present
STEAM Academics Association — *Bergen Community College*
- Refocused the club on STEM learning beyond the classroom and students' paths into four-year programs
- Introduced students to aerospace engineering through hobby rocketry
- Connected a student with an NJIT admissions counselor and fee waiver; they enrolled there in Fall 2026

Awards

First Place, STEM Research Center Poster Competition, Bergen Community College — August 2026
Leadership Award, Office of Student Life, Bergen Community College — April 2026
Qualified for the VEX U World Championship, REC Foundation — March 2026
Judges Award, MCCC Annual VEX U Qualifier — January 2026

Technical Skills

- **Design & CAD**: Onshape, Fusion 360, Mechanical Product Design
- **Fabrication**: Rapid Prototyping, FDM / SLA 3D Printing, Soldering, Mechanical Assembly
- **Electronics**: Circuit Design, Breadboarding, Microcontrollers
- **Software**: C/C++, Python, Git